DL Based Image Caption Generator

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Guide

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OUTLINE

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Abstract

- Image Caption Generator is a Deep Learning-based system that automatically generates meaningful textual descriptions for images.
- This project uses the Flickr8K dataset to train a model that understands image content and produces natural language captions.
- The system combines Convolutional Neural Networks (CNN) for image feature extraction and Recurrent Neural Networks (LSTM) for sequence generation.
- The model learns visual patterns and language structure simultaneously to generate human-like captions. This project demonstrates the integration of Deep Learning and Natural Language Processing techniques.

Problem Statement

With the rapid growth of digital images on the internet, manually describing images is time-consuming and inefficient. There is a need for an automated system that can:

- Understand the content of an image
- Identify objects and activities of a image
- Generate meaningful and grammatically correct captions

Existing traditional image processing methods cannot understand semantic meaning. Therefore, an intelligent Deep Learning-based system is required to generate accurate captions automatically.

Dataset Description

- The Flickr8K Dataset is a benchmark dataset widely used for image caption generation tasks in Deep Learning.
- It contains 8,000 real-world images, each paired with five different human-written captions, making it suitable for training models that combine Computer Vision and Natural Language Processing.

Dataset Structure

- Images Folder
- Captions File

Proposed Solution

The proposed system uses a hybrid Deep Learning architecture:

- CNN for extracting image features.
- LSTM (Long Short-Term Memory) network for generating captions.
- Trained on Flickr8K Dataset containing 8000 images with 5 captions per image.

Proposed System Steps:

- Image preprocessing and feature extraction using pretrained CNN
- Text preprocessing (tokenization, padding, vocabulary building)
- Training LSTM model using image features + text sequences
- Generating captions
- This model learns to predict the next word in a sequence based on image features and previously generated words.

Text Preprocessing

- **Tokenization**: Tokenization converts **words into numbers** because neural networks cannot understand text directly.
- **Padding:** Padding makes all sequences equal length.
- **LSTM:** LSTM is a special type of RNN used for **sequence data** like:

i. Sentences

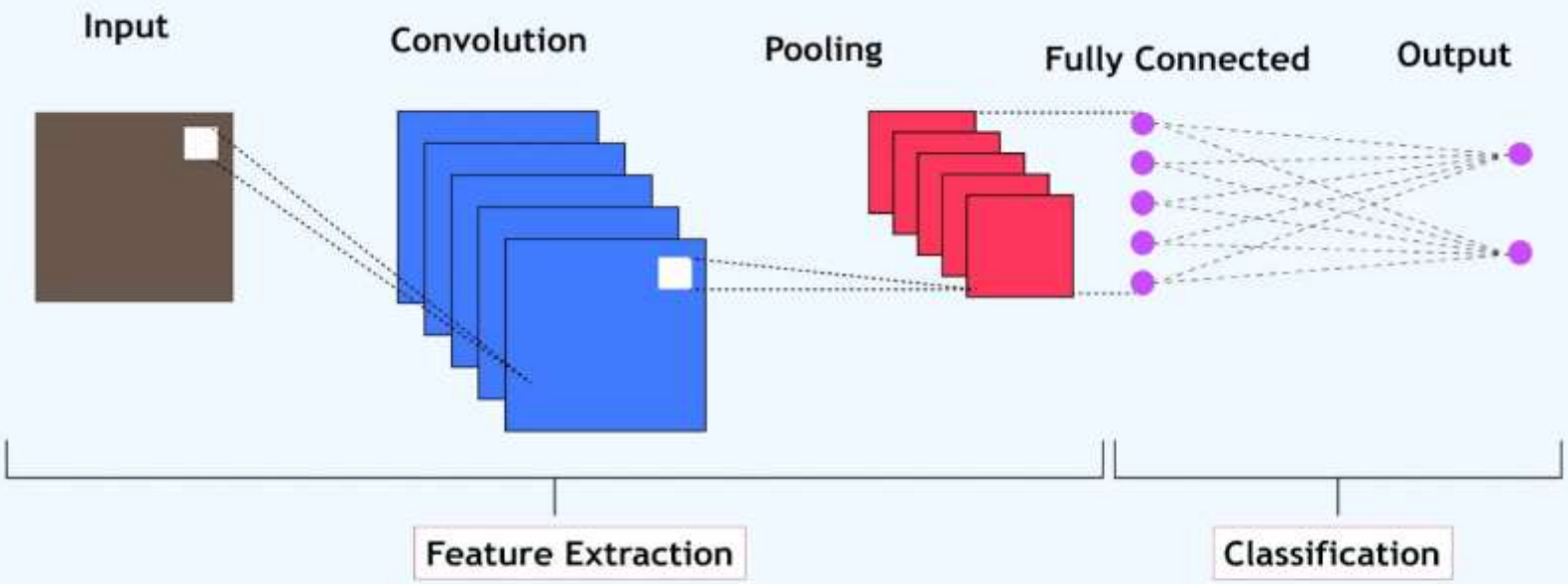
ii. Time series

iii. Speech

iv. Captions

- We uses LSTM for **generating captions word by word**

The Architecture of Convolutional Neural Networks



Convolution Neural Network

- A **Convolutional Neural Network (CNN)** is a deep learning model
- 1 **Input Layer** – The image is given as input to the network.
- 2 **Convolution Layer** – Small filters slide over the image to extract features like edges, textures, and patterns.
- 3 **ReLU Activation** – Applies f(x)=max(0,x) to introduce non-linearity and remove negative values.
- 4 **Pooling Layer** – Reduces the size of feature maps (usually using Max Pooling) to decrease computation and overfitting.
- 5 **Flatten Layer** – Converts the 3D feature maps into a 1D vector.
- 6 **Fully Connected + Output Layer** – Processes features and gives final prediction using Softmax (multi-class) or Sigmoid (binary).

System Architecture

CNN Feature Extractor

CNN extracts image feature vector.

Text Processing Module

Embedding layer converts words into dense vectors.

LSTM Decoder

Image features + word embeddings merged.

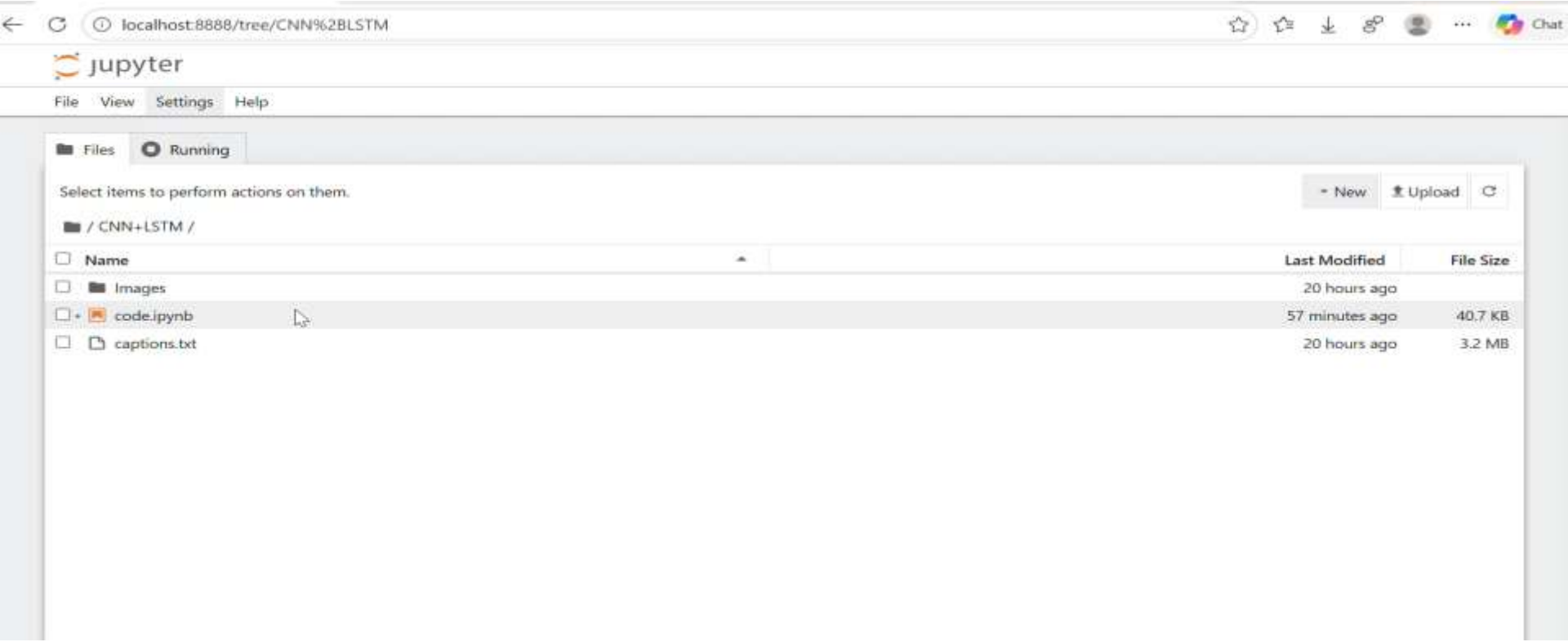
LSTM processes combined input.

Caption Generator

Dense + Softmax layer predicts next word.

Generates full sentence word-by-word

Video of Project Demo



Evaluation Metrics

BLEU Score (Bilingual Evaluation Understudy)

- BLEU-1(72%), BLEU-2(68%), BLEU-3(64%), BLEU-4(59%) measure n-gram overlap.
- Compares generated caption with reference captions.
- Higher BLEU score → better caption quality.

Insights & Recommendations

Insights :

- Feature Extraction Works Very Well
- Text Preprocessing Strongly Impacts Results
- Sequence Length Matters
- Vocabulary Size Affects Model Complexity

Recommendations :

- Use Larger Dataset
- Replace LSTM with Transformer
- Improve Evaluation Metrics

Conclusion

The Deep Learning-based Image Caption Generator successfully combines Computer Vision and NLP techniques to generate meaningful captions for images.

The model trained on Flickr8K dataset can generate human-like descriptions with reasonable accuracy. This project demonstrates how CNN and LSTM networks work together to understand visual data and convert it into natural language.

Future Scope

The project can be improved by:

- Using larger datasets like MS COCO
- Implementing Transformer-based models
- Using Attention Mechanism for better accuracy
- Deploying as a web application
- Integrating with assistive technologies for visually impaired users

Thank You